Required output for each member

Each member should complete **two separate RRL analysis entries** using the same documentation format.

Required column	Information to extract
1. Author(s) and Year	Complete APA 7 reference
2. Research Objective	Problem investigated
3. Algorithm / Model	Methods proposed, implemented, or reviewed
4. Dataset	Source, size, and characteristics
5. Evaluation Metrics	Performance measurements used
6. Key Findings	Actual research results
7. Limitations / Research Gap	Identified weaknesses or limitations
8. Relevance to AYUDA	How the study supports our system
9. Implementation Decision	Adopt, adapt, benchmark, or exclude, with justification

Important: Members should read the actual papers rather than relying solely on abstracts or AI-generated summaries. For systematic reviews, document the methods reviewed rather than claiming the authors implemented all those algorithms.

Level 3 — Supporting recruitment-management research

Assigned member: Sam · Three studies

These papers are intended to justify Modules 1 and 4, particularly candidate search, rediscovery, and workflow management. Sam does not need to identify a new AI algorithm for each standard module.

RRL 13 — Searching for Talent: The Information Retrieval Challenges of Recruitment Professionals

Russell-Rose & Chamberlain · Business Information Review · 2016

This study examines how recruitment professionals search for candidates, formulate and refine search queries, and assess results.

Tony Russell-Rose, Jon Chamberlain, 2016

What Sam should investigate: Search criteria, retrieval difficulties, query refinement, candidate availability, and the importance of up-to-date candidate information.

AYUDA relevance: Supports the candidate rediscovery module and the need for searchable, reusable talent records.

Read RRL 13

Searching for Talent: The Information Retrieval Challenges of Recruitment Professionals

Author(s) and Year	Russell-Rose, T., & Chamberlain, J. (2016). Searching for talent: The information retrieval challenges of recruitment professionals. Business Information Review. https://doi.org/10.1177/0266382116631849. The uploaded author manuscript does not specify the final volume, issue, or page range.
Research Objective	To investigate the information-seeking behavior, search strategies, query

	formulation techniques, result evaluation practices, and search-system requirements of recruitment professionals. The study addresses the limited understanding of recruiters' complex information retrieval needs and how existing search tools support their candidate sourcing activities.
Algorithm / Model	Survey-based Information Retrieval (IR) analysis. The researchers conducted a 40-question online questionnaire covering recruiter demographics, search tasks, query formulation, result evaluation, and ideal search-system features. They examined the use of Boolean queries, synonyms, query expansion, filtering, and relevance ranking. The paper also discussed attribute-based searching and search-by-example as potential improvements, but did not implement or evaluate these as new algorithms.
Dataset	The study analyzed 64 valid responses from recruitment professionals collected between June 9 and August 1, 2015. The survey initially received 416 responses, of which 69 were complete; five were excluded because of contradictory or nonsensical answers. Respondents were recruited through professional social media groups and SurveyMonkey Audience's North American HR panel. The survey contained 40 questions across five sections.
Evaluation Metrics	The study used descriptive survey statistics, including response percentages, medians, and weighted average scores on five-point Likert scales. Measurements included search task duration, number of queries submitted, query formulation time, time spent assessing results, frequency of search criteria, and perceived importance of search features. It did not report machine-learning metrics such as accuracy, precision, recall, F1-score, or NDCG.
Key Findings	Candidate search was reported by 76% of respondents, while 52% used proprietary internal databases. The median candidate search task took 3 hours, involved 5 queries, and required 5 minutes per query to formulate. Boolean logic (4.25/5), synonyms (4.16/5), and query expansion (4.02/5) were highly valued. Recruiters primarily filtered results by job function (4.34/5) and location (4.29/5). The recency of retrieved candidate information received the highest search-management importance score (4.21/5). The study also found that recruiters frequently refined searches and evaluated results until they identified a sufficient number of suitable candidates.
Limitations / Research Gap	The study relied on a relatively small sample of 64 recruiters and self-reported survey responses, which may not fully represent actual search behavior. The authors acknowledged possible demographic and recruitment-method biases. Since the number of professionals reached was unknown, the survey participation rate could not be calculated. The study did not implement or experimentally compare proposed search improvements, nor did it evaluate candidate retrieval accuracy or operational efficiency using actual recruitment-system logs.
Relevance to AYUDA	The study directly supports AYUDA's proposed candidate search and rediscovery module by identifying practical recruitment search challenges. Its findings provide a basis for searchable candidate records, structured filters for job function and location, synonym-aware search, reusable search criteria, and relevance-based candidate ranking. The emphasis on information recency also supports tracking when candidate profiles and availability were last verified, helping recruiters assess whether previously stored candidates remain suitable for new vacancies.

Implement Decision	Adapt. AYUDA should adapt the study's user-centered information retrieval principles by providing structured candidate search, relevant filters, reusable searches, and indicators of candidate-information recency. Synonym expansion and search-by-example can be considered for further evaluation. The study should be used as a requirements and workflow-design reference, rather than a benchmark for machine-learning performance, because it does not present a tested matching algorithm.
Synthesis	Russell-Rose and Chamberlain (2016) investigated the information retrieval challenges experienced by recruitment professionals through a survey of 64 respondents. The study found that candidate sourcing involves complex and iterative search processes, with recruiters relying on Boolean queries, synonyms, query expansion, and multiple information sources to identify suitable candidates. Results showed that a typical search task required a median of three hours and five queries, while job function, location, and the recency of candidate information were important factors in evaluating search results. The researchers also identified limitations in existing recruitment search tools, particularly their support for query formulation and refinement. The study provides a practical foundation for AYUDA's proposed candidate search and rediscovery module by emphasizing the need for efficient retrieval of existing candidate records. Its findings support structured search filters, reusable search criteria, synonym-aware retrieval, and indicators of candidate information recency. However, the study relied on self-reported survey data and did not experimentally evaluate an improved retrieval system. Therefore, AYUDA can adapt its user-centered search principles while validating the proposed features against the client's actual recruitment workflow and candidate database.

RRL 14 — Understanding Recruiters' Information Seeking Behavior in Talent Search

Kaya & Bogers · ACM CHIIR · 2023

This paper investigates recruiter search behavior and the ways recruiters formulate, refine, and evaluate searches during talent sourcing.

ir.webis.de+1

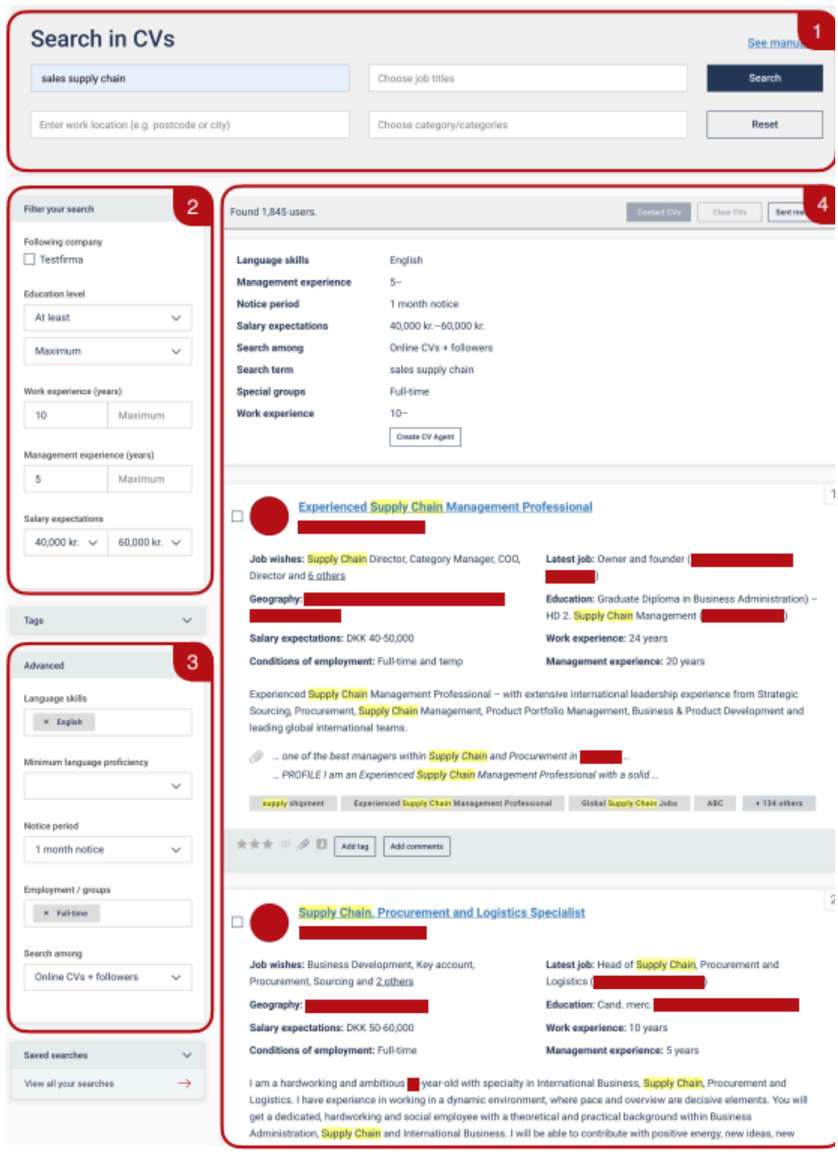
What Sam should investigate: Search sessions, information retrieval methods, filtering, recruiter workload, and how recommendations could be incorporated into the existing workflow.

AYUDA relevance: Helps justify integrating candidate rediscovery and matching recommendations into a recruiter-controlled search process.

Read RRL 14

Understanding Recruiters' Information Seeking Behavior inTalent Search

Author(s) and Year	Kaya, M., & Bogers, T. (2023). Understanding recruiters' information seeking behavior in talent search. *Conference on Human Information Interaction and Retrieval*, 14–23. https://doi.org/10.1145/3576840.3578311 Kaya and Bogers (2023)

Research Objective	The study investigated recruiters' information-seeking behavior, focusing on search sessions, query refinement, filtering practices, and candidate evaluation. It examined how search behavior changes across recruitment tasks and how search characteristics relate to candidate response rates.
Algorithm / Model	 Figure 1: The interface of the CV search engine. Personal information has been masked. The interface consists of four main components: (1) the search bars, which allow for keyword search, location-based search, and search for job titles and job categories; the regular (2) and advanced (3) filter panes; and (4) the search results, which contain teaser representations for each CV. Matching terms are automatically highlighted in yellow in the CV teasers in the results list.
Dataset	The study analyzed recruiter search logs from Jobindex, a Danish recruitment platform, covering September 2016 to November 2021. The original dataset contained 157,046 search tasks and approximately five million queries. After filtering, 128,133 sessions containing approximately four million queries remained, consisting of 92,020 matching tasks and 36,113 recruiting tasks.
Evaluation Metrics	The researchers measured session duration, query count, query length, term diversity, advanced search operator usage, and recruitment-specific filter usage. Positive candidate response rate was used as a proxy for search success. ANOVA, Tukey's HSD tests, effect sizes, correlation analysis, and an independent-samples t-test were used to examine differences and relationships

	in the recorded data.
Key Findings	The study found that matching sessions averaged 27 queries and approximately 27 minutes, while recruiting sessions averaged 42 queries and approximately five hours and 28 minutes. Recruiters progressively formulated longer and more diverse queries and increased their use of advanced search operators and filters. Location was the most frequently used filter, appearing in approximately 88% of matching sessions and 81% of recruiting sessions. The overall positive candidate response rate was 12.8%, while search duration showed a negligible relationship with response rates. These findings demonstrate that talent search involves iterative refinement and substantial candidate assessment effort.

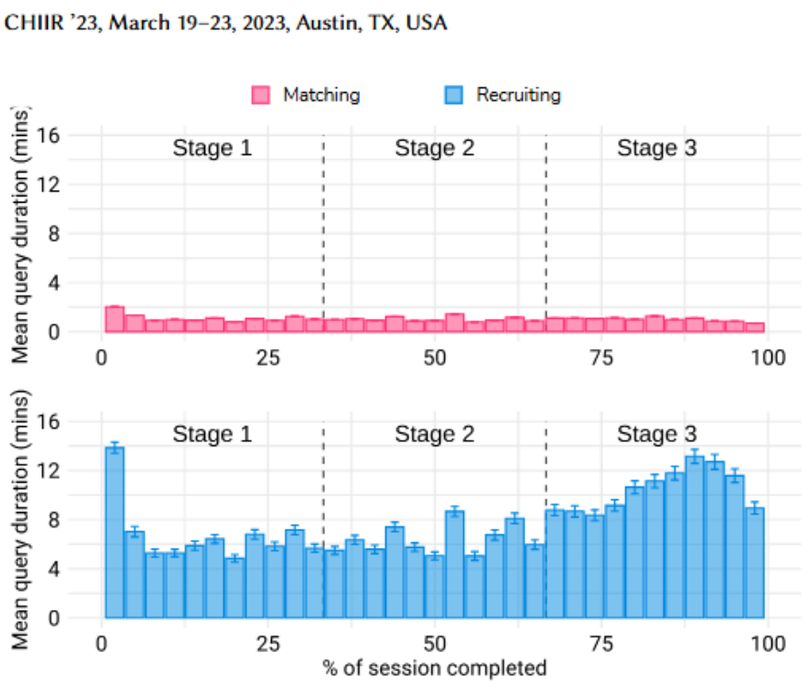


Figure 2: Mean query duration of queries issued at different(normalized) points in time during a search session for bothmatching jobs (top, N = 92,020) and recruiting jobs (bottom, N= 36,113). Both plots include the three search stages accordingto the partitioning described in Section 3.2.3.

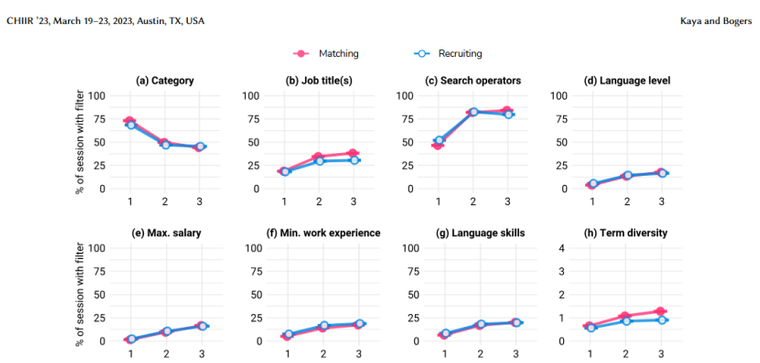
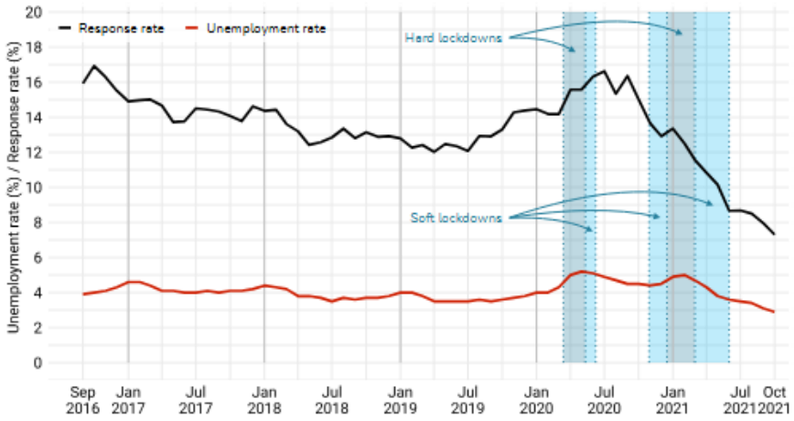


Figure 3: Changes in filter usage across the three different search stages for the eight metrics in Table 2 with the highest effectsize $\eta 2$. Error bars based on standard error are included, but are nearly invisible due to the low standard deviation and highsample size. With the exception of the difference in use of the job title filter (b) between recruiting stages 2 and 3, all means forthe three stages are statistically significant different for both matching and recruiting at $p < 0.001$ with multiple comparison onthe ANOVA results using the Tukey's

	HSD test.  Figure 4: Time series plot of response rate (black) and unem-ployment rate (red). All hard and soft lockdowns during theCovid-19 pandemic have been added as areas shaded in darkand light blue respectively
Limitations / Research Gap	The study relied on search logs from a single Danish recruitment organization and lacked detailed clickstream data, preventing the researchers from tracing individual queries to shortlisted candidates. Its division of search sessions into three stages was approximate, while positive response rates did not establish actual applications or hiring outcomes. Additionally, the study did not evaluate an AI-based recommendation system, leaving its potential effectiveness in supporting recruiter searches untested.
Relevance to AYUDA	The study supports AYUDA's proposed Candidate Sourcing and Talent Database by emphasizing the importance of iterative searching, advanced filtering, and efficient candidate information retrieval. Its findings provide a basis for developing flexible search functions that help recruiters locate and reconsider previously sourced candidates. The study also informs how AI-assisted recommendations could complement existing recruitment workflows while maintaining recruiter involvement in candidate evaluation.
Implement Decision	**Adapt and Benchmark.** AYUDA can adapt the study's keyword-searching, structured-filtering, and iterative-search principles for its centralized talent database. AI-assisted semantic retrieval may be investigated as an additional feature, although it was not evaluated in the study. The proposed approach should be benchmarked against conventional search methods using retrieval relevance, search completion time, and recruiter usability.
Synthesis	Kaya and Bogers (2023) examined recruiters' information-seeking behavior using 128,133 search sessions containing approximately four million queries from Jobindex. Their findings demonstrated that talent search involves repeated query refinement, advanced filtering, and substantial candidate assessment, with longer recruitment tasks requiring considerably more time. The study provides a research foundation for AYUDA's proposed Candidate Sourcing and Talent Database by supporting flexible information retrieval and iterative candidate searching. However, because the researchers did not evaluate AI-based recommendations, AYUDA must independently assess whether semantic retrieval and explainable decision support improve candidate rediscovery and recruitment efficiency.

RRL 15 — The impact of human resources information systems and business process management implementations on recruiting process performance: A case study

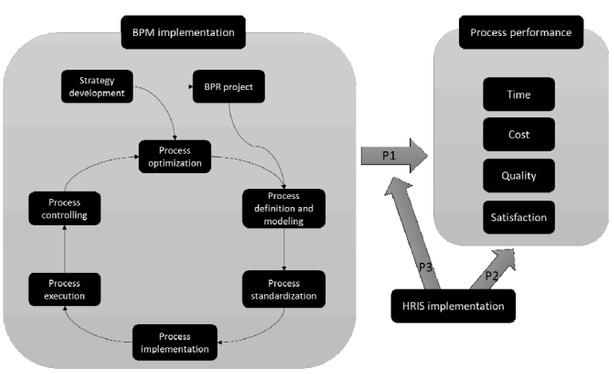
Laumer, Maier, & Eckhardt · Journal of Business Economics · 2014

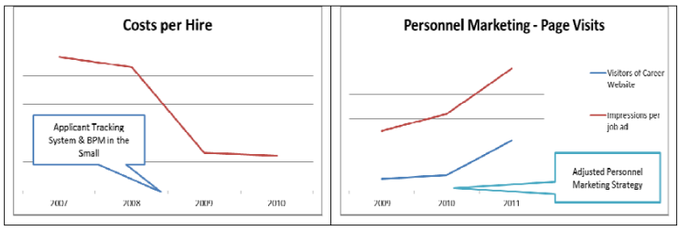
FAU CRIS

What Sam should investigate: Recruitment process measures, workflow monitoring, process improvement, cycle time, and the limitations of the empirical findings.

AYUDA relevance: Supports the standard recruitment workflow and monitoring functions without claiming that a Kanban board is an innovation.

The impact of human resources information systems and business process management implementations on recruiting process performance: A case study

Author(s) and Year	Laumer, S., Maier, C., & Eckhardt, A. (2014). *The impact of human resources information systems and business process management implementations on recruiting process performance: A case study.* In *Proceedings of the 20th Americas Conference on Information Systems (AMCIS).* Association for Information Systems. https://aisel.aisnet.org/amcis2014/AdoptionofIT/GeneralPresentations/15/
Research Objectives	The study aimed to examine how Business Process Management (BPM) and a Human Resources Information System (HRIS) can improve the recruitment process by reducing costs, shortening recruitment cycle time, improving process quality, and increasing stakeholder satisfaction.
Algorithm/Model	 **Figure 1: Research Model** The study used a research model that examines the effects of Business Process Management **(BPM)** implementation and Human Resources Information System **(HRIS)** implementation on recruitment process performance. BPM includes process modeling, standardization, implementation, execution, controlling, and optimization. Recruitment performance is evaluated based on time, cost, quality, and stakeholder satisfaction. The model also considers how BPM and HRIS work together to improve the recruitment process.
Data Set	The study did not use a conventional numerical dataset. It used a qualitative case study of **one Swiss financial service provider** that had implemented BPM and an HRIS for its recruitment process. The researchers collected data during the post-implementation phase, after the new process had been operating for at least one year. The data included interviews with project-management personnel, HR management, and affected

	employees, as well as project documents and process-controlling results. The interviews were recorded, transcribed, coded according to the research model, and cross-checked by the researchers.
Evaluation Metrics	The study evaluated recruitment process performance using **recruitment cost, recruitment cycle time, process quality, and stakeholder satisfaction**. It also examined process-related indicators, such as job-posting costs, cost per hire, and career-website visits. The relationships between BPM, HRIS implementation, and recruitment performance were analyzed using survey-based measures and **linear regression analysis**.
Key Findings	The study reported that the company reduced its total staff-recruitment cost by 50% between 2007 and 2010, particularly the costs of job postings and cost per hire. The company also observed an increase in career-website visits and changes in impressions per job advertisement after reducing offline job advertisements and increasing the use of online job boards. These findings show that HRIS and BPM can support cost reduction, recruitment monitoring, and data-based process improvement.  **Figure 2: Benefits of HRIS and BPM implementation** Figure 2 shows a decrease in cost per hire after the Applicant Tracking System and BPM were implemented. It also shows an increase in career-website visits and impressions per job advertisement after the personnel-marketing strategy was adjusted.
Limitations / Research Gap	The study was limited because it focused on only one organization, one information-system implementation, and only the recruitment process. Therefore, the findings may not apply to other organizations, systems, or business processes. Further research should examine different organizations and recruitment settings to determine whether the findings can be generalized.
Relevance to AYUDA	The study is relevant to AYUDA because it shows how BPM and HRIS can organize recruitment activities, monitor workflow progress, reduce delays, and track performance indicators such as process time, cost, quality, and satisfaction. These findings support AYUDA's proposed standardized recruitment workflow, follow-up monitoring, activity records, and recruiter decision-support functions.
Implement Decision	**Adapt.** AYUDA can adapt the study's BPM and HRIS principles for standardized recruitment stages, workflow monitoring, process documentation, follow-up tracking, and performance indicators. However, the approach must be customized for AYUDA's multi-client recruitment workflow and combined with candidate sourcing, résumé processing, explainable candidate-to-job matching, and recruiter verification.
Synthesis	Laumer et al. (2014) demonstrated that combining Business Process Management (BPM) with a Human Resources Information System (HRIS) can improve recruitment performance. The case study showed reduced recruitment costs, shorter process times, improved process transparency, and increased stakeholder satisfaction. However, the study focused on only one organization,

	one information system, and one recruitment process. For AYUDA, the study supports the use of standardized recruitment stages, workflow monitoring, process documentation, follow-up tracking, and performance indicators. AYUDA can adapt these principles to its multi-client recruitment workflow while extending them with candidate rediscovery, résumé processing, explainable candidate-to-job matching, and recruiter decision support.